ISE 631: The Linear Programming Formulation of Markov Decision Processes

Fall 2026

Main read: Adelman [1] **Additional read/background:** Puterman's book [2]

Context

A Markov decision process with a finite state space is a linear program. Bellman's equation for the optimal value function is roughly equivalent to

$$\min_{V} \sum_{s} \nu(s)V(s) \quad \text{s.t.} \quad V(s) - \alpha \sum_{s'} p(s' \mid s,a)V(s') \;\geq\; r(s,a) \quad \forall (s,a),$$

whose dual variables $x(s,a)$ turn out to be the discounted expected number of times action a is taken in state s, so the dual feasible region is the set of achievable actions and its constraints are flow conservation. Complementary slackness then recovers the optimal policy directly from the support of the optimal dual. The formulation above has one constraint per state–action pair and is therefore unusable at scale, and the project is focused on what to do in network revenue management.

Main paper, tentative plan, possible questions

Adelman [1] applies the machinery above to network revenue management. In this problem, a firm sells products that consume bundles of resources of finite capacity over a finite horizon, and must decide in real time which requests to accept. A state in the exact dynamic program encodes the vector of remaining capacities together with the time remaining. Thus, it is intractable. Industry practice had long relied instead on *bid prices* obtained from a deterministic linear program in which random demand is replaced by its expectation (this is usually known as the fluid formulation). Given a solution of this formulation, each resource is assigned a price, and a request is accepted when its revenue exceeds the total price of the resources it consumes. The weakness of that approach is that the prices are static: they do not vary with time or remaining inventory.

Adelman applies approximate linear programming with an affine value-function approximation, writing the value of a state as a time-dependent constant plus a time-dependent linear function of the remaining capacities. The resulting program yields bid prices that are functions of the remaining time and inventory, known as *dynamic* bid prices. Unfortunately, the constraint set of the approximate program remains exponentially large, and the paper solves it by column generation.

Tentative plan:

- Familiarize yourself with Markov decision processes [2].

- Familiarize yourself with the main paper [1], main methodology and solution approach. Let me know if you need more material about network revenue management.
- Relate the content of the main paper to the topics seen in class, e.g., how bid prices are computed (complementary slackness?).
- Recognize the main result in the paper and learn its proof technique.
- Numerically test the solution approach (e.g., deterministic LP, static and dynamic bid-price policies, etc) using synthetic instances.

Some questions (you should come up with your own questions as well):

- What is the benefit of using an approximate value function?
- Why does the resulting bound dominate the deterministic (fluid) formulation?
- What is the pricing/separation subproblem? Is it easy or hard to solve? Why?
- What is the resulting gap between the dynamic bid-price policy (lower bound), the MDP optimal value and the fluid formulation (upper bound)?
- If you were reviewing this work, what other comments would you provide to the author(s)? Any modeling considerations? Any different approach or benchmark?

References

[1] D. Adelman. Dynamic bid prices in revenue management. *Operations Research*, 55(4):647–661, 2007. https://pubsonline.informs.org/doi/10.1287/opre.1060.0368

[2] M. L. Puterman Markov decision processes: discrete stochastic dynamic programming *John Wiley & Sons, 2014*